A redesigned clinical LLM agent with better prompts, safer tools, and a lightweight **memory loop** boosts success from 70% to **91%** without memory and **98%** with memory on MedAgentBench, while generalizing to **300 new EHR tasks** at 89% success rate.

MedAgentBench v2: Improving Medical LLM Agent Design

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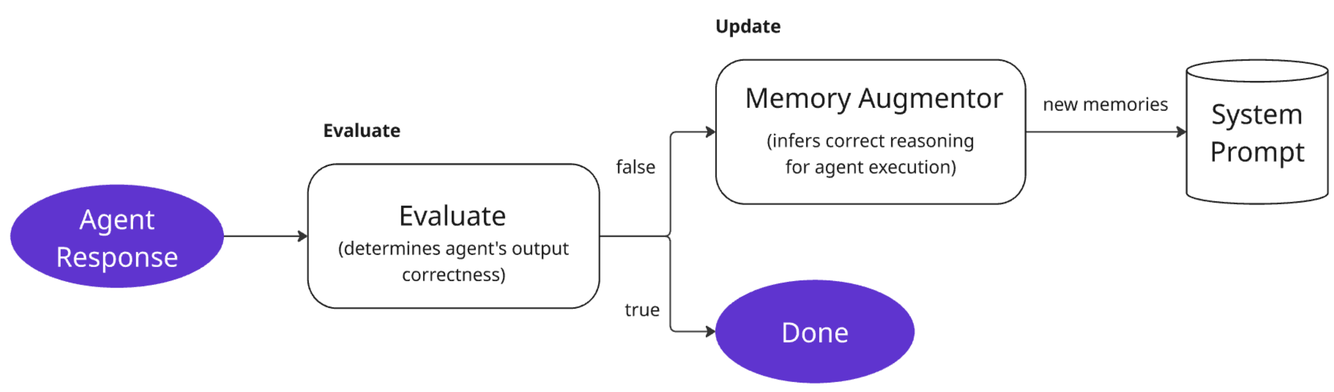

Figure 1. Memory-augmented agent loop: responses are evaluated, failures trigger a memory synthesis step that appends concise corrective instructions to the system prompt for future runs.

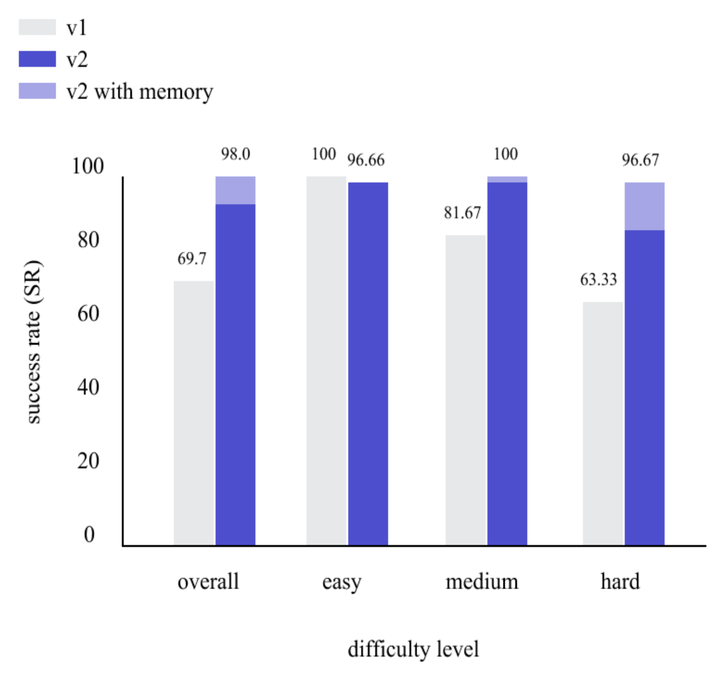

Figure 2. Success rate comparison: prior MedAgentBench v1 versus the redesigned agent before memory and after memory, showing gains with the memory component.

INTRODUCTION

- Clinical agents must operate reliably across multi-step EHR workflows, not just answer questions; targeted tool and prompt design markedly reduces failure modes that block safe deployment.
- A simple memory augmenter converts one-off errors into durable improvements, correcting conditional-logic mistakes and improving performance even on tasks without direct memories.

METHODS

- Base model: GPT-4.1; agent required to plan before any tool call and to validate preconditions prior to actions; outputs typed and minimal, with ISO dates where needed.
- Memory train/test: select five failed tasks per category to populate memory; test on held-out tasks to assess generalization of memory-guided improvements.
- New benchmark: 300 additional multi-step, clinically-derived tasks across imaging surveillance, medication validation, observation analysis, device monitoring, thyroid management, and immunization workflows; each with custom evaluators verifying clinical logic and valid FHIR resource fields.

RESULTS

- Overall SR: 91.0% without memory; 98.0% with memory on MedAgentBench tasks; hard-task SR rises to 96.67% with memory, including 100% on Tasks 9 and 10.
- Transfer effect: a memory crafted for HbA1C logic (missing result ≈ >1 year → place new order) improved potassium replacement behavior by correcting conditional bracketing, suggesting “style/vibe” learning.
- New-task generalization: 88.67% SR on 300 unseen, complex tasks, indicating robustness beyond the original benchmark.

DISCUSSION

- Operational semantics and data normalization remain pain points; tools and examples should teach EHR conventions, and physician-in-the-loop review is needed for consequential actions.
- Format brittleness in evaluators can mask correct reasoning; deployment should pair accuracy with governance which includes auditable memories, sandboxed writes, human review in combination with optimizing SR jointly with per-task cost and latency.

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- Ethics statement and COI information located at: https://stanfordmlgroup.github.io/projects/medagentbench/

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